The Physical Origin of the Refractive Field in Density Field Dynamics: Gravity as Electromagnetic Vacuum Loading

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We develop the interpretation that gravity arises from electromagnetic energy loading of the quantum vacuum, treated as a physical storage medium with vacuum force scale $K_0 = c^4/(8\pi G)$. The dimensionless scalar field ψ measures the fractional loading, and the vacuum refractive index $n = e^\psi$ follows uniquely from the multiplicative composition law via Cauchy's exponential functional equation. We interpret the full nonlinear DFD field equation as a stress–strain constitutive equilibrium of the vacuum medium, define gravitational strain and stress in analogy with continuum mechanics, and show that the vacuum exhibits reduced gravitational permittivity at low gradients—the mechanism behind flat rotation curves. The vacuum stiffness $K_0 = \hbar H_0^2/(8\pi\alpha^{57}c)$ is parameter-free given the master invariant and one measured scale. The constitutive split parameter $\kappa = \alpha/4 \approx 1.82\times10^{-3}$ is derived from gauge-emergence physics, not fitted. We quantify the EM–ψ coupling prediction, address the vacuum energy hierarchy within a framework consistent with the α^{57} suppression, and survey the 114-year intellectual lineage from Einstein's variable-c proposal through analog gravity to the present framework.

I. INTRODUCTION

The vacuum of quantum electrodynamics is not empty. It has a permittivity ε_0 and a permeability μ_0, setting the speed of light $c = 1/\sqrt{\varepsilon_0\mu_0}$. In the Density Field Dynamics (DFD) framework, these are not fixed constants of "empty space" but constitutive properties of a physical medium—the quantum vacuum—whose electromagnetic response is modified by the presence of mass-energy.

DFD reproduces all solar system tests of GR [17, 18], matches 175 galaxy rotation curves from the SPARC database [19], and derives $\alpha^{-1} = 137.036$ from Chern–Simons quantization [16]. The proton's rest mass is approximately 99% gluon field energy [20]—mass is concentrated energy localized within the vacuum. Lunar laser ranging constrains $|\dot{G}/G| < 10^{-13}$ yr^{-1} [21], consistent with a fixed vacuum stiffness.

This paper develops the *vacuum loading* interpretation of DFD, in which:

1. Mass deposits a fractional energy loading ψ in the vacuum, modifying its refractive index to $n = e^\psi$.

2. The vacuum resists deformation with a force scale (stiffness) $K_0 = c^4/(8\pi G)$, the same coefficient that appears in the Einstein field equations. The energy density of the loaded vacuum is $u_\psi = K_0\,|\nabla\psi|^2$.

3. The nonlinear DFD field equation emerges as the mechanical equilibrium condition for this loaded medium.

4. The vacuum stiffness is fixed by the internal topology plus one cosmological scale (H_0) through $K_0 = \hbar H_0^2/(8\pi\alpha^{57}c)$.

The paper is organized as follows. Section II defines and derives the vacuum stiffness K_0. Section III develops the constitutive relations. Section IV interprets the field equation as a constitutive equilibrium condition. Section V quantifies the EM–ψ coupling. Section VI presents the self-consistent argument for $n = e^\psi$. Section VII treats the nonlinear regime with stress–strain language. Section VIII relates DFD to prior work. Section IX discusses implications and open questions. The dimensional consistency of all key equations is verified in Appendix A.

II. THE VACUUM STIFFNESS K_0

A. Definition

The vacuum stiffness K_0 is a force scale (units of newtons) that converts the dimensionless ψ-field gradient into physical energy density:

$$u_\psi = K_0\,|\nabla\psi|^2, \qquad \boxed{K_0 \equiv \frac{c^4}{8\pi G} \approx 4.82\times10^{42}\ \text{N.}} \tag{1}$$

Since $[K_0] = \text{N}$ and $[|\nabla\psi|^2] = \text{m}^{-2}$, the energy density u_ψ has the correct units $\text{N}\cdot\text{m}^{-2} = \text{J/m}^3$.

This is the coefficient appearing in the Einstein field equations relating curvature to stress-energy. In DFD, it acquires a direct physical interpretation: K_0 is the *vacuum force scale* (stiffness) of the vacuum treated as an electromagnetic storage medium. Equivalently, $G = c^4/(8\pi K_0)$.

B. Expression via the master invariant

The DFD master invariant [16] states:

$$\frac{G\hbar H_0^2}{c^5} = \alpha^{57}. \tag{2}$$

Given the master invariant and one measured scale (H_0), K_0 is parameter-free:

$$K_0 = \frac{c^4}{8\pi G} = \frac{\hbar H_0^2}{8\pi\,\alpha^{57}\,c}. \tag{3}$$

This expresses K_0 entirely in terms of $(\hbar, c, H_0, \alpha)$ with no explicit G. Since α is derived from the $\mathbb{C}P^2 \times S^3$ topology and the exponent $57 = k_{\max} - N_{\text{gen}} = 60 - 3$ is topologically forced, the vacuum stiffness is topologically structured, with normalization set by the single cosmological scale H_0.

Equivalently, in Planck units:

$$K_0 = \frac{F_P}{8\pi}, \tag{4}$$

where $F_P = c^4/G$ is the Planck force.

C. The stiffness–topology chain

The internal volume $V_{\text{int}} = \text{Vol}(\mathbb{C}P^2) \times \text{Vol}(S^3) = 2\pi^2 \times 2\pi^2 = 4\pi^4$ enters the geometric prefactor $K_{\text{geom}} = \pi^{3/2}/24$, which determines α via the spectral action on $\mathbb{C}P^2 \times S^3$ [16]. The causal chain is:

$$V_{\text{int}} = 4\pi^4 \xrightarrow{K_{\text{geom}}} \alpha^{-1} = 137.036 \xrightarrow{\alpha^{57},\,H_0} G \;\to\; K_0. \tag{5}$$

The step $\alpha \to G$ requires H_0 as an external input (via the master invariant). The vacuum stiffness is exponentially sensitive to the internal topology: a 1% shift in α^{-1} produces a 57% shift in K_0.

III. CONSTITUTIVE RELATIONS

In the loading picture, the vacuum is modelled as a dielectric medium whose electromagnetic properties respond to the scalar field ψ. The constitutive relations are:

$$\epsilon(\psi) = \epsilon_0\, n(\psi)\, e^{+\kappa\psi}, \qquad \mu_{\text{mag}}(\psi) = \mu_0\, n(\psi)\, e^{-\kappa\psi}, \tag{6}$$

with $n = e^{\psi}$. The phase speed is preserved: $v_{\text{ph}} = 1/\sqrt{\epsilon\,\mu_{\text{mag}}} = c/n$.

The parameter κ controls the electric/magnetic split. At tree level, the Gordon optical metric gives $\kappa = 0$ (symmetric E/B response). The gauge-emergence auxiliary metric introduces a correction proportional to the effective coupling $\alpha_{\text{eff}} = \alpha/n_2^2 = \alpha/4$, where $n_2 = 2$ is the SU(2) frame stiffness from the $(3, 2, 1)$ partition. This gives the derived constitutive split:

$$\kappa = \frac{\alpha}{4} \approx 1.82 \times 10^{-3}. \tag{7}$$

This is the same value as the EM–ψ nonlinear threshold η_c (Eq. 9), confirming both arise from the same gauge-emergence physics. The product $\epsilon \cdot \mu_{\text{mag}} = \epsilon_0\mu_0\, e^{2\psi} = n^2/c^2$ is independent of κ.

A. Observables from the sector split

The modified Coulomb field near a massive body is:

$$E(r) = \frac{q}{4\pi\epsilon_0\, r^2}\, e^{-(1+\kappa)\psi(r)}. \tag{8}$$

Near the Sun, $\psi_\odot(R_\odot) \approx 4.2 \times 10^{-6}$, giving a fractional modification $\delta E/E \approx -(1+\kappa) \times 4.2 \times 10^{-6}$. The LPI violation parameter is $\xi(\kappa) = 1 - K_\epsilon^{(S)}\kappa + \mathcal{O}(\kappa^2)$, where $K_\epsilon^{(S)}$ is a species-dependent sensitivity.

B. The EM threshold

The threshold for EM back-reaction on ψ is determined by the gauge–ψ Lagrangian:

$$\eta_c = \alpha\,\sin^2\theta_W \approx \frac{\alpha}{4} \approx 1.8 \times 10^{-3}, \tag{9}$$

where $\eta \equiv U_{\text{EM}}/(\rho c^2)$. This value follows from the electroweak mixing angle projection onto the vacuum polarization vertex, as derived from the gauge–ψ Lagrangian coupling. We retain the statement that $\eta_c = \alpha/4$ arises from the gauge–ψ sector without claiming an independent microscopic derivation.

IV. THE FIELD EQUATION FROM VACUUM LOADING

A. Linear regime

In the Newtonian limit ($|\psi| \ll 1$, $\mu \to 1$), the vacuum response is linear and the field equation reduces to the Poisson equation:

$$\nabla^2\psi = -\frac{8\pi G}{c^2}\rho. \tag{10}$$

B. Constitutive analogy

The DFD field equation has the structure of a nonlinear flux-divergence equation:

$$\nabla\cdot\left[\mu\left(\frac{|\nabla\psi|}{a_*}\right)\nabla\psi\right] = -\frac{8\pi G}{c^2}(\rho - \bar{\rho}). \tag{11}$$

The vacuum loading picture interprets this as the equilibrium condition for a nonlinear medium. Define the gravitational strain and stress:

$$\text{Strain:} \quad s \equiv \frac{|\nabla\psi|}{a_*}, \tag{12}$$

$$\text{Stress:} \quad \boldsymbol{\sigma} \propto \mu(s)\,\nabla\psi, \tag{13}$$

where $a_* = 2a_0/c^2$ is the characteristic inverse-length scale and $\mu(s) = s/(1+s)$ is the DFD crossover function.

Then Eq. (11) takes the form $\nabla \cdot \boldsymbol{\sigma} \propto -\rho c^2$: the divergence of the vacuum stress balances the energy loading from matter.

The *structure* of this equation—nonlinear flux on the left, matter source on the right, with a field-dependent permittivity $\mu(s)$—is exactly that of a loaded nonlinear medium. The *coefficient* $8\pi G/c^2$ is not determined by the analogy alone; it is fixed by the DFD action principle [17], which specifies both the kinetic functional $W(|\nabla\psi|^2/a_*^2)$ and the matter–ψ coupling.

C. Energy density

The energy density of the ψ-field configuration is [17]

$$u_\psi = K_0\,|\nabla\psi|^2, \tag{14}$$

where $K_0 = c^4/(8\pi G)$. Since $[K_0] = \mathrm{N}$ and $[|\nabla\psi|^2] = \mathrm{m}^{-2}$, we have $[u_\psi] = \mathrm{J/m}^3$ as required. The prefactor K_0 and the absence of the elastic $\frac{1}{2}$ are both fixed by the action normalization that produces the correct field equation (11); they are not independently adjustable.

V. QUANTIFYING THE EM–ψ COUPLING

If the vacuum is truly a storage medium, EM fields must couple to the loading state: $\lambda \neq 1$, where λ controls EM back-reaction on ψ. This is testable via the SQMS Q-ratio:

$$\left.\frac{Q_{\text{low-f}}}{Q_{\text{high-f}}}\right|_{\text{DFD}} = 0.52 \pm 0.08, \qquad \left.\frac{Q_{\text{low-f}}}{Q_{\text{high-f}}}\right|_{\text{BCS}} = 3.60. \tag{15}$$

The sharp discrimination ($> 30\sigma$ separation) arises because in DFD with vacuum loading, the cavity's stored EM energy partially loads the vacuum at the ψ-mode frequency, and the frequency dependence of the ψ-channel loss inverts the Q-ratio relative to BCS theory.

Current constraints give $|\lambda - 1| \lesssim 3 \times 10^{-5}$ (accidental bound), with intentional search reach to $\sim 10^{-14}$. The SQMS Q-ratio test provides a shape measurement independent of the absolute value of $|\lambda - 1|$.

VI. WHY $n = e^\psi$: THE CAUCHY FUNCTIONAL EQUATION

A. The composition law

DFD postulates $n = e^\psi$ as the vacuum refractive index. The field equation yields $dn/d\psi = n$, whose unique solution with $n(0) = 1$ is $n = e^\psi$. But *why* should the vacuum's response be multiplicative rather than additive?

The answer lies in the physics of cumulative loading. Consider two successive loadings: first load the vacuum to state ψ_1, producing index $n(\psi_1)$. Then apply additional loading ψ_2 on top of the already-loaded vacuum. The total loading is $\psi = \psi_1 + \psi_2$ (loadings are additive in the field variable). The composition law

$$n(\psi_1 + \psi_2) = n(\psi_1) \cdot n(\psi_2) \tag{16}$$

—sequential loading of the vacuum is multiplicative—uniquely forces $n = e^\psi$ via Cauchy's exponential functional equation (assuming measurability).

Specifically, the functional equation $f(x + y) = f(x)f(y)$ with $f(0) = 1$ and f continuous and monotone has the unique solution $f(x) = e^{ax}$ for some $a \neq 0$ [22]. Setting $a = 1$ by the normalization convention $\psi = \ln n$ gives $n = e^\psi$.

The physical content is that there is no pre-existing medium with independent atomic structure. The vacuum *is* the medium, and ψ represents the *total* accumulated loading—not a perturbation on some fixed background. Unlike the Kerr effect in nonlinear optics (where $n = n_0 + n_2 I$ is additive because I is a small perturbation on a pre-existing medium with baseline n_0), the vacuum loading is cumulative and self-reinforcing: the medium through which the next increment of energy propagates has already been modified by all previous loading.

B. Why the linear ansatz fails

Suppose we tried $n = 1 + \beta\psi$ (additive, Kerr-like). Then:

$$n(\psi_1+\psi_2) = 1+\beta(\psi_1+\psi_2) \neq (1+\beta\psi_1)(1+\beta\psi_2). \tag{17}$$

The right side expands to $1+\beta\psi_1+\beta\psi_2+\beta^2\psi_1\psi_2$, which exceeds the left by $\beta^2\psi_1\psi_2$. The composition law fails.

Additionally: (i) for $\psi < -1/\beta$, the refractive index becomes negative, an unphysical regime; and (ii) the PPN parameters at second post-Newtonian order are wrong: $n^2 = 1 + 2\beta\psi + \beta^2\psi^2$ gives a ψ^2 coefficient of β^2, whereas GR requires 2 (twice the linear coefficient).

C. Three supporting arguments

Three independent lines of reasoning corroborate the composition law:

1. **Thermodynamic extensivity.** The loading ψ is an intensive thermodynamic variable. For a medium whose response depends only on the current state (not on the path), the partition function factorizes over independent loadings: $Z(\psi_1+\psi_2) = Z(\psi_1)\cdot Z(\psi_2)$. Since n is determined by Z, the multiplicative law follows.

2. **U(1) phase accumulation.** In the microsector, ψ modifies the gauge-field phase: $\phi = \int n\,ds = \int e^\psi\,ds$. The exponential ensures that phase accumulated over a path is the product of contributions

from each infinitesimal segment, the unique form compatible with the group structure of U(1) phase rotations.

3. **PPN consistency.** With $\psi = -2\Phi/c^2$ and $n = e^\psi$, the expansion $n^2 = e^{2\psi} = 1 + 2\psi + 2\psi^2 + \cdots$ automatically generates the correct PPN parameters $\gamma = 1$, $\beta = 1$ with all higher-order terms determined—no separate tuning at each PN order.

VII. THE NONLINEAR REGIME: STRESS–STRAIN INTERPRETATION

The DFD field equation (11) admits a natural interpretation in the language of continuum mechanics.

A. Definitions

The gravitational "strain" is the dimensionless gradient normalized by the saturation scale:

$$s = \frac{|\nabla\psi|}{a_*}, \tag{18}$$

where $a_* = 2a_0/c^2 \approx 2.67\times10^{-27}$ m^{-1} is the characteristic inverse-length scale and $a_0 = 2\sqrt{\alpha}\,cH_0$ is the crossover acceleration. Note that a_* has units of m^{-1} (not m/s^2), so the strain $s = |\nabla\psi|/a_*$ is indeed dimensionless.

The gravitational "stress" is defined, up to an overall constant, as the flux quantity whose divergence balances the source:

$$\boldsymbol{\sigma} \equiv K_0\,\mu(s)\,\nabla\psi, \tag{19}$$

where $K_0 = c^4/(8\pi G)$. This has dimensions $[\boldsymbol{\sigma}] = \mathrm{N}\cdot\mathrm{m}^{-1} = \mathrm{J/m}^2$, so $[\nabla\cdot\boldsymbol{\sigma}] = \mathrm{N/m}^2 = \mathrm{Pa}$, as verified in Appendix A.

The field equation (11) then reads $\nabla\cdot\boldsymbol{\sigma} = -\rho c^2$: the divergence of the vacuum stress balances the matter energy density. (The overall numerical coefficient is absorbed into K_0; see Sec. IV.)

These are physical interpretations of existing DFD quantities, not independent derivations.

B. Regime structure

The crossover function $\mu(s) = s/(1+s)$ produces two asymptotic regimes:

a. High strain ($s \gg 1$, Newtonian regime). $\mu(s) \to 1$, so $\sigma \propto \nabla\psi$ (linear, Hookean). The field equation reduces to $\nabla^2\psi = -(8\pi G/c^2)\rho$.

b. Low strain ($s \ll 1$, deep-field/MOND regime). $\mu(s) \approx s$, so $\sigma \propto |\nabla\psi|\nabla\psi$ (quadratic, reduced-permittivity regime). The field equation becomes:

$$\nabla\cdot\left[\frac{|\nabla\psi|}{a_*}\nabla\psi\right] = -\frac{8\pi G}{c^2}\rho. \tag{20}$$

C. Reduced gravitational permittivity at low gradients

The function $\mu(s)$ acts as a field-dependent *gravitational permittivity.* At high strain ($s \gg 1$), $\mu \to 1$ and the vacuum conducts gravitational flux at full Newtonian strength. At low strain ($s \ll 1$), $\mu \approx s \to 0$: the vacuum becomes a poor conductor of gravitational flux.

This is the physical mechanism behind flat rotation curves. By Gauss's law applied to the DFD field equation, the total gravitational flux through a sphere of radius r around a mass M is fixed:

$$\oint \mu\left(\frac{|\nabla\psi|}{a_*}\right)\nabla\psi\cdot d\mathbf{A} = -\frac{8\pi GM}{c^2}. \tag{21}$$

When μ is small (low gradients, large r), the gradient $|\nabla\psi|$ must be *larger* than the Newtonian value to carry the same flux. In the deep-field limit $\mu \approx s$, the flux becomes quadratic in $|\nabla\psi|$, and solving gives $|\nabla\psi| \propto 1/r$ rather than the Newtonian $1/r^2$. Since the acceleration $a = (c^2/2)|\nabla\psi|$, this yields $v^2 = ra = \text{const}$—a flat rotation curve.

The analogy is to a nonlinear dielectric whose permittivity drops at low field strengths. In such a material, the same enclosed charge produces a larger electric field at large distances than in a linear dielectric, because the reduced permittivity cannot screen the flux as effectively. The DFD vacuum behaves identically: reduced gravitational permittivity at low gradients amplifies the gravitational field relative to the Newtonian prediction, eliminating the need for dark matter.

VIII. RELATION TO PRIOR WORK

The idea that gravity might arise from or be described through the electromagnetic properties of the vacuum has a long history.

A. Historical foundations

Einstein [1] proposed in 1911 that the gravitational field modifies the speed of light: $c(\Phi) = c_0(1 + \Phi/c^2)$. This is the earliest explicit treatment of gravity as a variable refractive index. DFD's $n = e^\psi$ is the nonlinear completion of Einstein's linear formula.

Wilson [2] proposed that mass creates a spatially varying dielectric constant in the vacuum, a direct precursor to DFD's constitutive-relation approach.

Gordon [3] showed that EM waves in a moving dielectric obey an effective "optical metric." DFD inverts this: the optical metric is fundamental, and the "moving dielectric" is the vacuum ψ-field.

Dicke [4] proposed that gravity could originate from modifications to the vacuum's electromagnetic properties, suggesting that the speed of light correlates with

the gravitational potential of the universe—the closest pre-DFD articulation of the vacuum loading concept.

B. Induced gravity and emergent approaches

Sakharov [5] proposed that gravity emerges from vacuum quantum fluctuations—"induced gravity." The Einstein–Hilbert action arises as a one-loop effective action from matter fields, providing a natural framework for DFD's claim that G (and hence K_0) is determined by vacuum mode structure.

Visser [10] reviewed Sakharov's programme, showing how G emerges from the spectrum of matter field fluctuations. The "stiffness" in Sakharov's picture is precisely $K_0 = c^4/(8\pi G)$.

C. Polarizable vacuum and optical approaches

de Felice [6] gave a systematic treatment of the gravitational field as an optical medium with position-dependent refractive index, deriving lensing, redshift, and Shapiro delay from the optical analogy. DFD elevates this from analogy to fundamental dynamics.

Puthoff [9] developed a "polarizable vacuum" model in which mass modifies the vacuum's permittivity and permeability via a scalar function, reproducing linearized GR. This is formally similar to DFD's constitutive relations, though DFD provides a dynamical action principle and microsector derivation.

Perlick [8] established the rigorous mathematical foundation for ray optics in curved spacetime via Fermat's principle.

D. Analog gravity and vacuum energy

Volovik [11] demonstrated that superfluid ^{3}He-A provides an exact analog of gravity, electromagnetism, and the Standard Model as collective modes of the quantum vacuum. Crucially, Volovik showed that the vacuum energy computed from mode sums is cancelled by the equilibrium condition—directly supporting DFD's distinction between K_0 (vacuum force scale) and ρ_Λ (residual strain), and anticipating the α^{57} suppression mechanism.

Barceló, Liberati, and Visser [13] provided a comprehensive review of analog gravity models. Electromagnetic analogs (dielectric media with gradient refractive index) are discussed as gravitational simulators. DFD makes the inverse claim: the gravitational ψ-field *is* the physical refractive medium, not merely an analogy.

E. Spectral geometry

Connes and Chamseddine [7, 12] developed the spectral action framework from which DFD's α derivation [16] descends. The spectral action on a product geometry $M^4 \times F$ simultaneously produces the Einstein–Hilbert gravitational action and the Standard Model gauge kinetic terms. DFD identifies F with $\mathbb{C}P^2 \times S^3$ and uses specific heat-kernel coefficients to derive $\alpha^{-1} = 137.036$.

Recent work by García-Compean et al. [14] and De Rham et al. [15] extends the optical-gravity analogy into the metamaterial regime.

IX. DISCUSSION

A. The vacuum energy hierarchy

The vacuum loading picture distinguishes three scales:

Scale	Value	Role
QFT cutoff $\rho_P c^2$	$\sim 10^{113}$ J/m^3	Naive mode-sum
Stiffness K_0	$\sim 10^{42}$ N	Vacuum force scale
Dark energy $\rho_\Lambda c^2$	$\sim 10^{-9}$ J/m^3	Residual strain

The critical distinction: K_0 is the vacuum force scale (resistance to deformation), not an energy density. The observed $\rho_\Lambda \sim 10^{-9}$ J/m^3 is the residual loading, a "strain" of order $H_0^2/c^2 \sim 10^{-52}$ m^{-2}. The vacuum loading picture provides a natural framework for understanding the vacuum energy hierarchy, consistent with the α^{57} suppression: the physical microsector has only $k_{\max} = 60$ modes, and the residual vacuum energy is suppressed by $\alpha^{57} \sim 10^{-122}$ relative to the Planck scale, rather than being set by a continuum mode sum up to M_P.

This parallels Volovik's argument [11] that in superfluid ^{3}He, the vacuum energy from mode sums vastly overestimates the gravitational effect, because most of the vacuum energy is cancelled by the equilibrium condition.

B. The vacuum stiffness and the master invariant

Given the master invariant $G \cdot \hbar \cdot H_0^2/c^5 = \alpha^{57}$ [16] and one measured scale, K_0 is parameter-free: $K_0 = \hbar H_0^2/(8\pi\alpha^{57}c)$. Whether this expression admits an independent derivation from the microsector topology remains open.

C. The chain of reasoning

The vacuum loading interpretation can be stated as a single chain:

1. The vacuum is a storage medium with ε_0, μ_0.
2. Mass loads the medium: $\psi = \ln n$.
3. The composition law forces $n = e^\psi$ (Cauchy).

4. The stiffness is topologically structured, with normalization set by H_0: $K_0 = \hbar H_0^2/(8\pi\alpha^{57}c)$.
5. The cosmological constant is residual strain.
6. QFT overestimates by overcounting modes.
7. EM–ψ coupling is predicted and testable.

D. Experimental programme

The key predictions are:

- SQMS Q-ratio: 0.52 ± 0.08 (DFD) vs. 3.60 (BCS).
- LPI slope $\xi(\kappa)$: species-dependent if $\kappa \neq 0$.
- TE/TM cavity asymmetry: $\sim 2\kappa\psi$ frequency splitting, with $\kappa = \alpha/4$ giving $\sim 10^{-3}\psi$ level splitting.
- Polarization rotation through $\nabla\psi$ gradients.

E. Open questions

1. The exact value of $\lambda - 1$ requires computing the EM–ψ mode overlap on $\mathbb{C}P^2 \times S^3$.
2. The sector-split parameter $\kappa = \alpha/4$ is derived from the gauge-emergence auxiliary metric; experimental confirmation via TE/TM splitting would test this prediction.
3. Time dependence of K_0: if $G(t)$ varies with cosmic epoch, implications for nucleosynthesis need quantification.
4. Whether the vacuum loading picture and Sakharov's programme can be unified through a single spectral action calculation.

X. CONCLUSIONS

We have developed the vacuum loading interpretation of DFD, in which the gravitational ψ-field arises from fractional electromagnetic energy loading of the quantum vacuum. The exponential refractive law $n = e^{\psi}$ follows uniquely from the multiplicative composition law via Cauchy's functional equation, supported by three independent arguments (thermodynamic extensivity, U(1) phase, PPN consistency). The vacuum stiffness $K_0 = \hbar H_0^2/(8\pi\alpha^{57}c)$ is determined by topology plus one scale measurement. The nonlinear regime admits a continuum-mechanics interpretation in which the vacuum exhibits reduced gravitational permittivity at low gradients—the mechanism behind flat rotation curves. The framework provides a natural understanding of the vacuum energy hierarchy consistent with the α^{57} suppression, and makes falsifiable predictions for the SQMS Q-ratio test and cavity–atom comparisons.

Appendix A: Dimensional Consistency of Key Equations

We verify the SI dimensional consistency of every key equation.

Quantity	**Result**	
$K_0 = c^4/(8\pi G)$	N	✓
$s = \lvert\nabla\psi\rvert/a_*$	dimensionless	✓
$\boldsymbol{\sigma} = K_0\mu\nabla\psi$	N/m	✓
$\nabla\cdot\boldsymbol{\sigma}$	Pa	✓
$\nabla\cdot(\mu\nabla\psi)$	m^{-2}	✓
$8\pi G\rho/c^2$	m^{-2}	✓
$u_\psi = K_0\lvert\nabla\psi\rvert^2$	$\mathrm{J/m}^3$	✓
$G\hbar H_0^2/c^5$	dimensionless	✓
$a_0 = 2\sqrt{\alpha}\,cH_0$	$\mathrm{m/s}^2$	✓
$a_* = 2a_0/c^2$	m^{-1}	✓
$\eta_c = \alpha\sin^2\theta_W$	dimensionless	✓
$\kappa = \alpha/4$	dimensionless	✓
ξ_{LPI}	dimensionless	✓
$k_a a_0^2 = \frac{3}{2}(cH_0)^2$	$\mathrm{m}^2/\mathrm{s}^4$	✓

All equations are dimensionally consistent in SI units.

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